

軟體測試與資安檢測

教師：王凡

授課時間：每週三 9：10—12：00

博理館 113

課程代號：EE 5170

識別碼：921 U9550

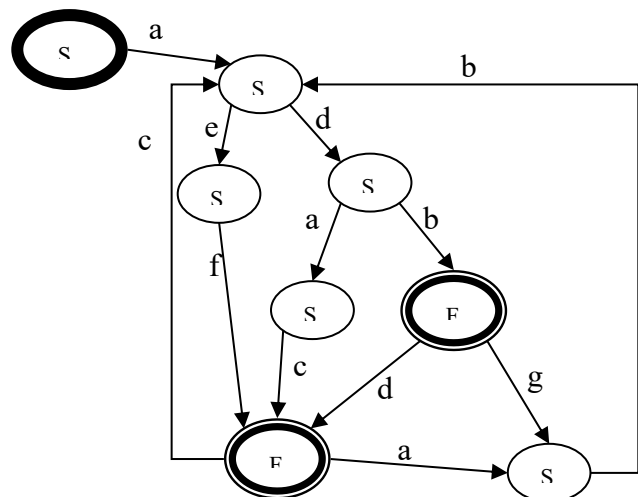
民國 110 年春季班

QUIZ (2021/04/14)

Open-book, open-internet, no discussion

Name: _____ Department : _____ Student ID: _____

1. We have the following finite-state automata. S0 is the initial state while F0 and F1 are the final states. Please write down the regular expression of the automata.



2. Continuing from problem 1, if each cycle in your regular expression is to be repeated up to 3 times, please calculate the number of test paths in this regular expression.

3. Continuing from problem 1 with the cost of each test action to the right. Please calculate the total cost of the test plan if each cycle in your regular expression is to be repeated up to 3 times.

Cost(a)= 1

Cost(b)=2

Cost(c)=3

Cost(d)=4

Cost(e)=5

Cost(f)=6

Cost(g)=7

4. Continuing from problem 1, please compute the minimum number of test paths to fulfill 100% edge coverage.

5. Suppose we have the following two predicates:

$$(a \ \&\& \ (x < 100 \rightarrow y \neq z)) \ \parallel \ !b$$

$$x < y + 10 \ \&\& \ !(\cos(z) < 0.5)$$

Please write down two test cases for predicate coverage.

6. Continuing from problem 5, please write down two test cases for atom coverage.

7. Continuing from problem 5, please write down test cases for combinatorial coverage.